

CHARLES MAXIMILLIAN THORN

(551) 775-7047 | cthorn@stevens.edu | linkedin.com/in/charles-thorn-2491573a1

EDUCATION

Stevens Institute of Technology | Hoboken, NJ

Expected May 2029

Bachelor of Engineering, Mechanical Engineering

GPA: 3.5+ | **Honors:** Dean's List (Jan 2026), MAC Fall Sports Academic Honor Roll (Jan 2026)

Coursework: Engineering Statics (ENGR 211), Thermodynamics (ME 234), Circuits & Systems (ENGR 245), Differential Equations, Engineering Design I & II (ENGR 111, 122), Linear Algebra & Vectors, Multivariable Calculus I

SKILLS

Programming: Arduino IDE (proficient), C++ (basic), Python (basic), Java (basic)

Tools: SolidWorks (proficient), ChatGPT (proficient), Claude (proficient), FDM 3D Printing (design, slicing, fabrication)

PROJECTS

Robot Arena | ENGR 122

Jan. 2026 – May 2026

SolidWorks, Arduino, FDM 3D Printing

- Designed robot main housing in SolidWorks, engineered a unique frame optimized to exceed spec for minimum weight and maximum structural strength.
- Optimized pathfinding between checkpoints using ultrasonic feedback and ArUco marker localization.
- Ranked first in preliminary standings and advanced to the finals of the Gallois Autonomous Robot Competition, 1 of 12 teams selected from roughly 650 first-year engineers.

Automatic Plant Watering System | ENGR 111

Sep. 2025 – Dec. 2025

Arduino IDE, SolidWorks

- Led SolidWorks design of an automated system that monitors soil moisture and transmits sensor readings wirelessly to the Stevens network.
- Led Arduino programming to read soil moisture sensor data and drive automatic watering controls.
- Iterated enclosure design through multiple revisions to optimize for 3D print time, structural strength, and material efficiency.

WORK EXPERIENCE

Stevens Institute of Technology — MOCORE Lab | Hoboken, NJ

Jul. 2026 – Present

Undergraduate Student Researcher (Raviraj Nataraj, PhD)

- Engineered a dual inflatable TPU bladder system and custom 3D-printed trough housing for a forearm restraint jig used in VR-based motor rehabilitation research.
- Resolved heat-weld corner seam geometry and pressurized inflation circuitry across successive fabrication iterations.
- Built two functional prototypes currently under evaluation.

Blue Line Pool Management | Exclusive Pool Management | Ridgefield & Mahwah, NJ

Jul. 2024 – Present

Lifeguard

- Supervise pool operations independently at luxury residential properties and a commercial aquatic facility, enforcing health and safety compliance protocols.
- Apply American Red Cross CPR, AED, and emergency response training to incident prevention and rescue readiness.

LEADERSHIP & EXTRACURRICULARS

American Institute of Aeronautics and Astronautics (AIAA) | Hoboken, NJ

Sep. 2025 – Present

Secretary (AY 2026–2027)

- Record and distribute official meeting minutes, design presentation materials for chapter meetings and events, and build Google Forms to coordinate member availability and event scheduling.

SAE Aero Design Club | Hoboken, NJ

Sep. 2025 – Present

Structures Subteam Member

Stevens RockSAT-C, NASA Student Launch | Hoboken, NJ

Sep. 2026 – Present

Vehicle Team Member

Stevens Institute of Technology — Track & Field | Hoboken, NJ

Sep. 2025 – Present

NCAA Div III Student-Athlete, Cross Country & Track (Winter, Spring)

Memberships: AIAA, ASME, National Society of Black Engineers (NSBE), Stevens Commuter Student Union